Go Programming — Weeks 1 to 7

Week 1

GCD and LCM of two given numbers.

```
package main

import "fmt"

func main() {
    var a, b int
    fmt.Print("Enter two numbers: ")
    fmt.Scan(&a, &b)

    x, y := a, b
    for y != 0 {
        x, y = y, x%y
    }

    gcd := x
    lcm := (a * b) / gcd

    fmt.Println("GCD =", gcd)
    fmt.Println("LCM =", lcm)
}
```

Week 2

Centered pyramid of numbers.

```
package main

import "fmt"

func main() {
    n := 5
    num := 1

    for i := 1; i <= n; i++ {
        for space := 1; space <= n-i; space++ {
            fmt.Print("  ")
        }

        for j := 1; j <= i; j++ {
            fmt.Printf("%d ", num)
            num++
        }
        fmt.Println()
    }
}
```

Week 3

Using a struct imported from another package.

```
// person/student.go
package person

type Student struct {
    Name string
    Age  int
}

// main.go
package main

import (
    "fmt"
    "weeks/person"
)

func main() {
    s := person.Student{
        Name: "Hero",
        Age:  0,
    }

    fmt.Println(s.Name)
}
```

Week 4

Calculate standard deviation using the math package.

```
package main

import (
    "fmt"
    "math"
)

func main() {
    nums := []float64{2, 4, 4, 4, 5, 5, 7, 9}

    mean := 0.0
    for _, n := range nums {
        mean += n
    }
    mean /= float64(len(nums))

    sum := 0.0
    for _, n := range nums {
        sum += math.Pow(n-mean, 2)
    }

    sd := math.Sqrt(sum / float64(len(nums)))
    fmt.Println("Standard deviation =", sd)
}
```

Week 5

Floyd's triangle.

```
package main

import "fmt"

func main() {
    n := 3
    num := 1

    for i := 1; i <= n; i++ {
        for j := 1; j <= i; j++ {
            fmt.Print(num, " ")
            num++
        }
        fmt.Println()
    }
}
```

Week 6

Input and concatenation of two strings.

```
package main

import "fmt"

func main() {
    var a, b string

    fmt.Print("Enter first string: ")
    fmt.Scan(&a)

    fmt.Print("Enter second string: ")
    fmt.Scan(&b)

    fmt.Println("Addition of strings:", a+b)
}
```

Week 7

Check whether a string is a palindrome.

```
package main

import (
    "fmt"
    "strings"
)

func main() {
    var s string
    fmt.Print("Enter a string: ")
    fmt.Scan(&s)

    s = strings.ToLower(s)
    r := ""

    for i := len(s) - 1; i >= 0; i-- {
        r += string(s[i])
    }

    if s == r {
        fmt.Println("Palindrome")
    } else {
        fmt.Println("Not Palindrome")
    }
}
```